Suppose longitudinal linkage is available: each student $i$ in subgroup $S$ has a prior quantile $U_i$ *and* a current quantile $V_i$. With these linked pairs, we can compute the true latent conditional percentiles

$$P_i = F_0(V_i \mid U_i),$$

where $F_0$ is the conditional CDF from the baseline copula $C_0$. The *truth benchmark* for the subgroup is

$$\theta_S^{\text{true}} = 100\,\mathbb{E}(P_S).$$

This is the quantity we would report if individual-level linkage existed. Panel A shows the density $f_{H_S}(p)$ of the true growth regime alongside the bridge-recovered density $f_{\hat{H}_S}(p)$. The vertical lines mark their respective means.

The fundamental role of Panel A is to *define what accuracy means*: accuracy can only be assessed relative to a well-defined truth. Without a linked-data benchmark, "accuracy" becomes internal consistency rather than genuine error measurement.[1]

The small displacement between the amber (true) and black (bridge) densities is the *bridge cost*: the price paid for erasing individual-level linkage and relying on the copula bridge instead.